1. Project Overview

1.1 Project Name

SRP8-044 – Rear Suspension

1.2 Lead

Tessa Sitjes

1.3 Scope

This project encompasses the development of the rear suspension geometry as well as the spring and damper system on SR8. This development includes optimising the subsystem's performance and the SR8 project's vision while complying with all necessary regulations.

1.4 Out of Scope

The project excludes the front suspension and the design of uprights, wheel hubs, and braking systems, although many aspects and subsystems are similar between the two.

1.5 Assemblies

- ASSY – REAR DAMPER LH
- ASSY – REAR DAMPER RH
- ASSY – REAR LOWER WISHBONE LH
- ASSY – REAR LOWER WISHBONE RH
- ASSY – REAR TOELINK LH
- ASSY – REAR TOELINK RH
- ASSY – REAR TOP WISHBONE LH
- ASSY – REAR TOP WISHBONE RH

2. Interfacing Requirements (PMS)

2.1 REAR DAMPERS

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type
ASSY - REAR WHEEL	Ensure full damper travel without interference with the wheel or tyre, maintaining suspension geometry and safe clearance under all operating conditions	

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type

2.2 REAR LOWER WISHBONES

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type
ASSY - REAR WHEEL	Must provide accurate mounting interface for the upright/wheel assembly and withstand transmitted loads from tyre contact forces	
ASSY - REAR QUARTER PANEL	Must ensure sufficient clearance to avoid interference throughout full suspension travel (bump, rebound, roll)	

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type
ASSY - OCCUPANT CELL TUB	Must provide structurally adequate mounting points to transfer suspension loads from the wishbones without excessive deformation or failure	

2.3 REAR TOELINKS

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type
ASSY - REAR WHEEL	Must provide a compatible mounting interface with the upright to enable accurate toe control and transmit lateral and longitudinal loads	
ASSY - REAR QUARTER PANEL	Must ensure sufficient clearance for toelink geometry and articulation throughout full suspension travel without interference	

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type

2.4 REAR TOP WISHBONES

2.1 Dependencies (what this project relies on)

Assembly	Requirement	Type

ASSY - REAR WHEEL	Must provide accurate mounting interface for the upright/wheel assembly and withstand transmitted loads from tyre contact forces	
ASSY - REAR QUARTER PANEL	Must ensure sufficient clearance to avoid interference throughout full suspension travel (bump, rebound, roll)	

2.2 Impacts (what relies on this project)

Assembly	Requirement	Type
ASSY - OCCUPANT CELL TUB	Must provide structurally adequate mounting points to transfer suspension loads from the wishbones without excessive deformation or failure	

3. ADR Requirements

I'm unsure if there may be other requirements relating to suspension, but the ones assigned apparently apply to trucks, trailers and multi-axle vehicles rather than SR8 which has independent suspension (I believe) and does not use load-sharing suspension systems like trucks.

Pulled directly from the ADR database.

ADR	Clause	Requirement	Verification category (for systems)
43	8.1	All the ‘Axles’ in an ‘Axle Group’ other than a ‘Twin Steer Axle Group’ must be related to each other through a ‘Load Sharing Suspension’ except:	Out of scope/Not applicable
43	8.1.1	This requirement must not apply to a ‘Close Coupled Axle Group’ on any motor vehicle or trailer (including a ‘Semi-trailer’) with a ‘GVM’ or ‘ATM’ less than 4.5 tonne provided that the load carrying capacity of each of the ‘Axles’ and the wheel and tyres fitted to each ‘Axle’ is at least 120% of the nominal load imposed on that ‘Axle’ with the vehicle or trailer at its ‘GVM’ or ‘ATM’;	Out of scope/Not applicable
43	8.1.2	This requirement must not apply to a ‘Close Coupled Axle Group’ fitted with a ‘Retractable Axle’ on any motor vehicle or trailer (including a ‘Semi-trailer’) with a ‘GVM’ or ‘ATM’ less than 4.5 tonne provided that the load carrying capacity of each other ‘Axle’	Out of scope/Not applicable

		and the wheels and tyres fitted to it is at least equal to the load imposed on that ‘Axle’ with the motor vehicle or trailer at its ‘GVM’ or ‘ATM’ and with the ‘Retractable Axle’ retracted	